Exercises 1

Exercise 1 (Random operation)**.** *Write down the linear transformation that with probability q_0 leaves a probabilistic bit alone and with probability q_1 inverts it. Is this the most general operation you can perform on a probabilistic bit?*

Exercise 2 (Guessing the coin)**.** *Let A and B be biased coins whose probability distributions are $(1/3, 2/3)$ and $(3/5, 2/5)$. With probability $1/2$ you are given a sample from coin A and with probability $1/2$ a sample from coin B. Your goal is to guess from which coin the sample came. What is the best strategy for doing this and with what probability does it succeed?*

Exercise 3 (Converting between representations)**.** *Let z be a complex number. Derive formulas for converting between the exponential representation $z = re^{i\varphi}$ and the algebraic representation $z = a + ib$.*

1. *Assume $z = re^{i\varphi}$ and express a and b in terms of r and φ.*
2. *Assume $z = a + ib$ and express r and φ in terms of a and b.*

Exercise 4 (Angle sum identities)**.** *Use Euler's formula and $e^{i(\alpha+\beta)} = e^{i\alpha} \cdot e^{i\beta}$ to derive a formula for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$.*

Exercise 5 (Multiplication geometrically)**.** *What is the geometric interpretation of the equation $r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1+\varphi_2)}$ for complex number multiplication?*

Exercise 6 (Inverse)**.** *In quantum computing we generally do not need to divide complex numbers. However, in case you are curious and would like to know how to do this, notice that $x/z = x \cdot 1/z$. Now it only remains to figure out how to find the inverse $1/z$.*

1. *Let $z = re^{i\varphi}$. Compute $1/z$ in the exponential form.*
2. *Show that the inverse of any $z \in \mathbb{C}$ can be found as follows: $1/z = \bar{z}/|z|^2$.*

Exercise 7 (Discriminating $|+\rangle$ and $|-\rangle$)**.** *Consider the qubit states $|+\rangle := \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle := \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Find a qubit operation U such that measuring the states $U|+\rangle$ and $U|-\rangle$ in the standard basis produces opposite outcomes with certainty, thus allowing the states to be perfectly discriminated.*

Exercise 8 (Bloch vector is a unit vector)**.** *Let $\psi_0, \psi_1 \in \mathbb{C}$ be such that $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$ is a qubit state, i.e., $|\psi_0|^2 + |\psi_1|^2 = 1$. Let (x, y, z) be the Bloch vector corresponding to $|\psi\rangle$:*

$$\begin{cases} x = 2\,\mathrm{Re}(\bar{\psi}_0\psi_1), \\ y = 2\,\mathrm{Im}(\bar{\psi}_0\psi_1), \\ z = |\psi_0|^2 - |\psi_1|^2. \end{cases}$$

Verify that $x^2 + y^2 + z^2 = 1$, meaning that (x, y, z) is a unit vector in $\mathbb{R}^3$.

Exercise 9 (Locating states on the Bloch sphere)**.** *Find the Bloch vectors corresponding to the following six qubit states:*

$$|0\rangle, \qquad |+\rangle := (|0\rangle + |1\rangle)/\sqrt{2}, \qquad |+i\rangle := (|0\rangle + i|1\rangle)/\sqrt{2},$$
$$|1\rangle, \qquad |-\rangle := (|0\rangle - |1\rangle)/\sqrt{2}, \qquad |-i\rangle := (|0\rangle - i|1\rangle)/\sqrt{2}.$$